

R, Databases & SQL

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Contents

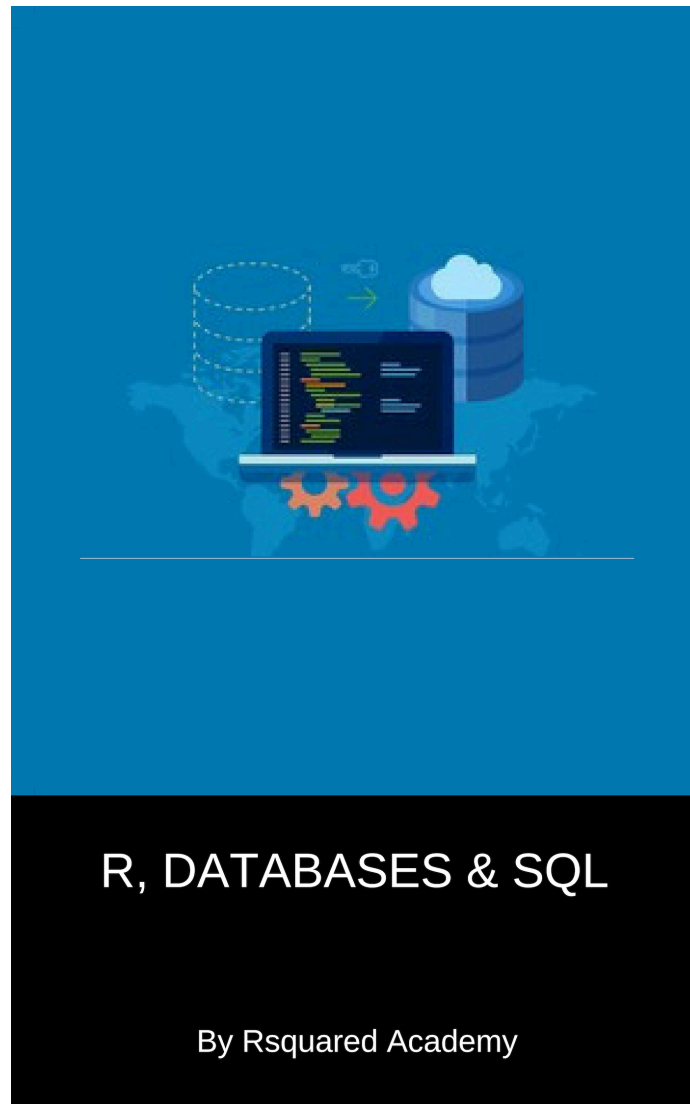
Preface	2
Structure of the book	3
Software information	3
About the Author	4
1 DBI	5
1.1 Introduction	5
1.2 Connection	5
1.2.1 Connection Summary	5
1.2.2 List Tables	6
1.2.3 List Fields	6
1.3 Querying Data	6
1.3.1 Entire Table	6
1.3.2 Few Rows	7
1.3.3 Read Data in Batches	7
1.4 Query	8
1.4.1 Query Status	8
1.4.2 Query Info	8
1.4.3 Latest Query	8
1.4.4 Rows Fetched	8
1.4.5 Rows Affected	9
1.4.6 Column Info	9
1.5 Create Table	9
1.6 Append Data	10
1.7 Insert Rows	11
1.8 Remove Table	12
1.9 SQLite Data Type	12
1.10 Close Connection	12
2 dbplyr	12
2.1 Introduction	12
2.2 Connect to Database	12
2.3 Reference Data	13
2.4 Query Data	13
2.5 Show Query	14
2.6 Collect Data	15

3	SQL Basics	15
3.1	Introduction	15
3.2	Set Up	16
3.3	Select Columns	16
3.3.1	Syntax	16
3.3.2	Select Single Column	17
3.3.3	Select Multiple Columns	17
3.3.4	Select All Columns	18
3.4	Limit	18
3.5	Distinct	19
3.6	Filter	19
3.7	And, Or & Not	21
3.8	BETWEEN	22
3.9	IN	23
3.10	IS NULL	24
3.11	LIKE	24
3.12	Exercises	25
4	SQL Advanced	26
4.1	Introduction	26
4.2	Set Up	26
4.3	Aggregate	26
4.4	Alias	28
4.5	Order By	28
4.6	Group By	30
4.7	Exercises	31
5	JOINS	31
5.1	Introduction	31
5.2	Set Up	31
5.3	INNER JOIN	32
5.4	LEFT JOIN	33
5.5	The SQLite RIGHT / FULL Trap	33
5.6	JOINS with dbplyr	34
5.7	semi_join and anti_join	35
5.8	Exercises	36
6	DBI Cheat Sheet	36
6.1	Connect, query, disconnect	36
6.2	Write data	37
6.3	Inspect a query	37
6.4	dbplyr in 4 verbs	37

Preface

Last verified: 2026-09-26 · R 4.5.2 · DBI · dbplyr 2.6.0 · RSQLite 2.4.6 · duckdb.

New here? Grab the one-page DBI cheatsheet, then work Chapters 1–5 in order — runnable code for each chapter lives in [code/](#).



Structure of the book

Chapter Chapter 1 introduces the **DBI** package. Chapter Chapter 2 explores data wrangling within database using the **dplyr** package. Chapters Chapter 3 and Chapter 4 introduce basic and advanced SQL. Chapter Chapter 5 combines tables with JOINS in SQL and dbplyr.

Software information

The R session information when compiling this book is shown below:

```
sessionInfo()
```

```
R version 4.4.2 (2024-10-31)
```

```
Platform: x86_64-pc-linux-gnu
```

```
Running under: Ubuntu 24.04.5 LTS
```

```
Matrix products: default
```

```
BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
```

```
LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.26.so;
```

```
LAPACK version 3.12.0
```

```
locale:
```

```
[1] LC_CTYPE=C.UTF-8      LC_NUMERIC=C          LC_TIME=C.UTF-8
[4] LC_COLLATE=C.UTF-8    LC_MONETARY=C.UTF-8   LC_MESSAGES=C.UTF-8
[7] LC_PAPER=C.UTF-8      LC_NAME=C             LC_ADDRESS=C
[10] LC_TELEPHONE=C        LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
```

```
time zone: UTC
```

```
tzcode source: system (glibc)
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

```
loaded via a namespace (and not attached):
```

```
[1] compiler_4.4.2 fastmap_1.2.0 cli_3.6.6      tools_4.4.2
[5] htmltools_0.5.9 rmarkdown_2.32 knitr_1.52     jsonlite_2.0.0
[9] xfun_0.61       digest_0.6.39 rlang_1.3.0    evaluate_1.0.5
```

We do not add prompts (`>` and `+`) to R source code in this book, and we comment out the text output with two hashes `##` by default, as you can see from the R session information above. This is for your convenience when you want to copy and run the code (the text output will be ignored since it is commented out). Package names are in bold text (e.g., **rmarkdown**), and function names are followed by parentheses (e.g., `bookdown::render_book()`). The double-colon operator `::` means accessing an object from a package.

About the Author

Aravind Hebbali is the founder of Rsquared Academy. He earned his Masters in Economics from Madras School of Economics. As an active R user, he has authored several R packages such as

- `olsrr`
- `rfm`
- `descriptr`
- `blorr`
- `xplorerr`

In 2015, he founded Rsquared Academy, a free and open source education initiative with focus on data science and analytics. Apart from self paced online courses, Rsquared Academy offers customized learning modules for corporates and universities.

You can find him on GitHub.

1 DBI

1.1 Introduction

In this chapter, we will learn to:

- connect to a SQLite database from R
- display database information
- list tables in the database
- query data
 - read entire table
 - read few rows
 - read data in batches
- create table in database
- overwrite table in database
- append data to table in database
- remove table from database
- generate SQL query
- close database connection

We will use the following R packages:

- DBI
- RSQLite

All the data sets used in this chapter can be found [here](#) and code can be downloaded from [here](#) or run from `code/ch1-dbi.R` in this repo.

1.2 Connection

The first step is to connect to a database. We will connect to an in memory SQLite database using `dbConnect()`.

```
con <- dbConnect(RSQLite::SQLite(), ":memory:")
```

NOTE: We use SQLite here because it needs zero setup — no server to install. The exact same DBI commands work with DuckDB, a high-speed analytical engine, for larger analytical queries (see R4DS Ch. 21 at scale):

```
# install.packages("duckdb") # once; high-speed analytical engine
con_duck <- dbConnect(duckdb::duckdb())
dbWriteTable(con_duck, "ecom", ecom)
dbGetQuery(con_duck, "SELECT referrer, AVG(duration) FROM ecom GROUP BY referrer")
```

1.2.1 Connection Summary

We can get the more information about the connection using `summary()`.

```
summary(con)
```

Length	Class	Mode
1	SQLiteConnection	S4

1.2.2 List Tables

Now that we are connected to a database, let us list all the tables present in it using `dbListTables()`.

```
dbListTables(con)
```

```
[1] "ecom" "sqlite_stat1" "sqlite_stat4"
```

1.2.3 List Fields

Time to explore the `ecom` table in the database. Use `dbListFields()` to list all the fields in the table.

```
dbListFields(con, "ecom")
```

```
[1] "referrer" "device" "bouncers" "n_visit" "n_pages" "duration"
```

1.3 Querying Data

The main objectives of connecting to a database are to:

- query data from the tables already present
- create new tables
- overwrite existing tables
- delete existing tables

Let us begin with querying data. We can do this in the following ways:

- read an entire table at once
- read few rows from a table
- read data in batches

1.3.1 Entire Table

We can read an entire table from a database using `dbReadTable()`.

```
dbReadTable(con, 'ecom')
```

	referrer	device	bouncers	n_visit	n_pages	duration
1	google	laptop	1	10	1	693
2	yahoo	tablet	1	9	1	459
3	direct	laptop	1	0	1	996
4	bing	tablet	0	3	18	468
5	yahoo	mobile	1	9	1	955
6	yahoo	laptop	0	5	5	135
7	yahoo	mobile	1	10	1	75
8	direct	mobile	1	10	1	908
9	bing	mobile	0	3	19	209
10	google	mobile	1	6	1	208
11	direct	laptop	1	9	1	738
12	direct	tablet	0	6	12	132

13	direct	mobile	0	9	14	406
14	yahoo	tablet	0	5	8	80
15	yahoo	mobile	0	7	1	19
16	bing	laptop	1	1	1	995
17	bing	tablet	0	5	16	368
18	google	tablet	1	7	1	406
19	social	tablet	0	7	10	290
20	social	tablet	0	2	1	28

In some cases, we may not need the entire table but only a specific number of rows. Use `dbGetQuery()` and supply a SQL statement specifying the number of rows of data to be read from the table. In the below example, we read ten rows of data from the `ecom` table.

1.3.2 Few Rows

```
dbGetQuery(con, "select * from ecom limit 10")
```

	referrer	device	bouncers	n_visit	n_pages	duration
1	google	laptop	1	10	1	693
2	yahoo	tablet	1	9	1	459
3	direct	laptop	1	0	1	996
4	bing	tablet	0	3	18	468
5	yahoo	mobile	1	9	1	955
6	yahoo	laptop	0	5	5	135
7	yahoo	mobile	1	10	1	75
8	direct	mobile	1	10	1	908
9	bing	mobile	0	3	19	209
10	google	mobile	1	6	1	208

In case of very large table, we can read data in batches using `dbSendQuery()` and `dbFetch()`. We can mention the number of rows of data to be read while fetching the data using the query generated by `dbGetQuery()`.

1.3.3 Read Data in Batches

```
query <- dbSendQuery(con, 'select * from ecom')
result <- dbFetch(query, n = 15)
result
```

	referrer	device	bouncers	n_visit	n_pages	duration
1	google	laptop	1	10	1	693
2	yahoo	tablet	1	9	1	459
3	direct	laptop	1	0	1	996
4	bing	tablet	0	3	18	468
5	yahoo	mobile	1	9	1	955
6	yahoo	laptop	0	5	5	135
7	yahoo	mobile	1	10	1	75
8	direct	mobile	1	10	1	908
9	bing	mobile	0	3	19	209

10	google	mobile	1	6	1	208
11	direct	laptop	1	9	1	738
12	direct	tablet	0	6	12	132
13	direct	mobile	0	9	14	406
14	yahoo	tablet	0	5	8	80
15	yahoo	mobile	0	7	1	19

1.4 Query

1.4.1 Query Status

To know the status of a query, use `dbHasCompleted()`. It is very useful in cases of queries that might take a long time to complete.

```
dbHasCompleted(query)
```

```
[1] FALSE
```

1.4.2 Query Info

`dbGetInfo()` will return the following:

- the sql statement
- number of rows fetched
- number of rows modified/affected
- status of the query

```
dbGetInfo(query)
```

```
$statement
```

```
[1] "select * from ecom"
```

```
$row.count
```

```
[1] 15
```

```
$rows.affected
```

```
[1] 0
```

```
$has.completed
```

```
[1] FALSE
```

1.4.3 Latest Query

To get the latest query, use `dbGetStatement()`.

```
dbGetStatement(query)
```

```
[1] "select * from ecom"
```

1.4.4 Rows Fetched

To check the number of rows of data returned by a query, use `dbGetRowCount()`.

```
dbGetRowCount(query)
```

```
[1] 15
```

1.4.5 Rows Affected

To know the number of rows modified or affected in the table, use `dbGetRowsAffected()`.

```
dbGetRowsAffected(query)
```

```
[1] 0
```

1.4.6 Column Info

To know the name of the columns and their data types, use `dbColumnInfo()`.

```
dbColumnInfo(query)
```

	name	type
1	referrer	character
2	device	character
3	bouncers	integer
4	n_visit	double
5	n_pages	double
6	duration	double

Always clear the result set once you are done fetching and inspecting it.

```
dbClearResult(query)
```

1.5 Create Table

So far we have explored querying data from an existing table. Now, let us turn our attention to creating new tables in the database.

1.5.0.1 Introduction

To create a new table, use `dbWriteTable()`. It takes the following 3 arguments:

- connection name
- name of the new table
- data for the new table

```
x <- 1:10
y <- letters[1:10]
trial <- tibble::tibble(x, y)
dbWriteTable(con, "trial", trial)
```

Let us check if the new table has been created.

```
dbListTables(con)
```

```
[1] "ecom" "sqlite_stat1" "sqlite_stat4" "trial"
```

```
dbExistsTable(con, "trial")
```

```
[1] TRUE
```

Let us query data from the new table.

```
dbGetQuery(con, "select * from trial limit 5")
```

	x	y
1	1	a
2	2	b
3	3	c
4	4	d
5	5	e

1.5.0.2 Overwrite Table

In some cases, you may want to overwrite the data in an existing table. Use the `overwrite` argument in `dbWriteTable()` and set it to `TRUE`.

```
x <- sample(100, 10)
y <- letters[11:20]
trial2 <- tibble::tibble(x, y)
dbWriteTable(con, "trial", trial2, overwrite = TRUE)
```

Let us see if the **trial** table has been overwritten.

```
dbGetQuery(con, "select * from trial limit 5")
```

	x	y
1	38	k
2	57	l
3	67	m
4	9	n
5	5	o

1.6 Append Data

You can append data to an existing table by setting the `append` argument in `dbWriteTable()` to `TRUE`.

```
x <- sample(100, 10)
y <- letters[5:14]
trial3 <- tibble::tibble(x, y)
dbWriteTable(con, "trial", trial3, append = TRUE)
```

Let us quickly check if the new data has been appended to the **trial** table.

```
dbReadTable(con, "trial")
```

	x	y
1	38	k
2	57	l
3	67	m
4	9	n
5	5	o
6	56	p

```

7 84 q
8 80 r
9 27 s
10 36 t
11 24 e
12 78 f
13 52 g
14 49 h
15 41 i
16 51 j
17 88 k
18 99 l
19 21 m
20 44 n

```

We can also use `sqlAppendTable()` to append data to an existing table.

```

sqlAppendTable(con, "ecom", head(ecom))

<SQL> INSERT INTO `ecom`
  (`referrer`, `device`, `bouncers`, `n_visit`, `n_pages`, `duration`)
VALUES
  ('google', 'laptop', TRUE, 10, 1, 693),
  ('yahoo', 'tablet', TRUE, 9, 1, 459),
  ('direct', 'laptop', TRUE, 0, 1, 996),
  ('bing', 'tablet', FALSE, 3, 18, 468),
  ('yahoo', 'mobile', TRUE, 9, 1, 955),
  ('yahoo', 'laptop', FALSE, 5, 5, 135)

```

1.7 Insert Rows

1.7.0.1 Introduction

We can insert new rows into existing tables using:

- `dbExecute()`
- `dbSendStatement()`

Both the function take 2 arguments:

- connection name
- sql statement

```

# use dbExecute
dbExecute(con,
  "INSERT into trial (x, y) VALUES (32, 'c'), (45, 'k'), (61, 'h')"
)
## [1] 3

```

```
# use dbSendStatement
res <- dbSendStatement(con,
  "INSERT into trial (x, y) VALUES (25, 'm'), (54, 'l'), (16, 'y')"
)
dbClearResult(res)
```

1.8 Remove Table

If you want to delete/remove a table from the database, use `dbRemoveTable()`.

```
dbRemoveTable(con, "trial")
```

1.9 SQLite Data Type

If you want to know the data type, use `dbDataType()`.

```
dbDataType(RSQLite::SQLite(), "a")
```

```
[1] "TEXT"
```

```
dbDataType(RSQLite::SQLite(), 1:5)
```

```
[1] "INTEGER"
```

```
dbDataType(RSQLite::SQLite(), 1.5)
```

```
[1] "REAL"
```

1.10 Close Connection

It is a good practice to close connection to a database when you no longer need to read/write data from/to it. Use `dbDisconnect()` to close the database connection.

```
dbDisconnect(con)
```

2 dbplyr

2.1 Introduction

In this chapter, we will learn to query data from a database using `dplyr`.

We will use the following R packages:

- DBI
- RSQLite
- dbplyr
- dplyr

All the data sets used in this chapter can be found [here](#) and code can be downloaded from [here](#) or run from `code/ch2-dbplyr.R` in this repo.

2.2 Connect to Database

Let us connect to an in memory SQLite database using `dbConnect()`.

```
con <- dbConnect(RSQLite::SQLite(), ":memory:")
```

We will copy the `ecom` data to the database so that we can use it for running dplyr statements. We read the same `ecom` data set used across chapters (1000 rows, 8 columns including `country` and `purchase`) from the local `data/web.csv` snapshot, falling back to the online copy if needed.

```
web_path <- if (file.exists("data/web.csv")) "data/web.csv" else "https://raw.githubusercontent.com/rsquaredacademy/datasets/master/web.csv"
ecom <- readr::read_csv(web_path, show_col_types = FALSE)
ecom <- dplyr::select(ecom, referrer, device, bouncers, n_visit, n_pages, duration,
country, purchase)
dplyr::copy_to(con, ecom)
```

2.3 Reference Data

In order to use dplyr functions, we need to reference the table in the database using `tbl()`.

```
ecom2 <- dplyr::tbl(con, "ecom")
ecom2
```

```
# A query: ?? x 8
# Database: sqlite 3.53.3 [:memory:]
  referrer device bouncers n_visit n_pages duration country      purchase
  <chr>    <chr>    <int>   <dbl>   <dbl>   <dbl> <chr>          <int>
1 google  laptop      1      10      1      693 Czech Republic    0
2 yahoo   tablet      1       9      1      459 Yemen              0
3 direct  laptop      1       0      1      996 Brazil            0
4 bing    tablet      0       3     18      468 China              1
5 yahoo   mobile      1       9      1     955 Poland            0
6 yahoo   laptop      0       5      5     135 South Africa     0
7 yahoo   mobile      1      10      1      75 Bangladesh       0
8 direct  mobile      1      10      1     908 Indonesia         0
9 bing    mobile      0       3     19     209 Netherlands      0
10 google mobile      1       6      1     208 Czech Republic    0
# i more rows
```

2.4 Query Data

We will look at some simple examples. Let us start by selecting `referrer`, `device` and `duration` columns from `ecom2`.

```
select(ecom2, referrer, device, duration)

# A query: ?? x 3
# Database: sqlite 3.53.3 [:memory:]
  referrer device duration
  <chr>    <chr>    <dbl>
1 google  laptop     693
2 yahoo   tablet     459
3 direct  laptop     996
4 bing    tablet     468
```

```

5 yahoo    mobile    955
6 yahoo    laptop    135
7 yahoo    mobile    75
8 direct   mobile    908
9 bing     mobile    209
10 google  mobile    208
# i more rows

```

We can filter data as well. Filter all the rows from `ecom2` where `duration` is greater than 300.

```

filter(ecom2, duration > 300)

# A query: ?? x 8
# Database: sqlite 3.53.3 [:memory:]
  referrer device bouncers n_visit n_pages duration country purchase
  <chr>    <chr>    <int>   <dbl>   <dbl>   <dbl> <chr>    <int>
1 google   laptop      1      10      1      693 Czech Republic    0
2 yahoo    tablet      1       9      1      459 Yemen              0
3 direct   laptop      1       0      1      996 Brazil            0
4 bing     tablet      0       3     18      468 China              1
5 yahoo    mobile      1       9      1      955 Poland            0
6 direct   mobile      1      10      1      908 Indonesia          0
7 direct   laptop      1       9      1      738 Jamaica            0
8 direct   mobile      0       9     14      406 Ireland            1
9 bing     laptop      1       1      1      995 United States      0
10 bing     tablet      0       5     16      368 Peru               1
# i more rows

```

Time to do some grouping and summarizing. Let us compute the average time spent on the site for different types of devices.

```

ecom2 %>%
  group_by(device) %>%
  summarise(avg_duration = mean(duration))

# A query: ?? x 2
# Database: sqlite 3.53.3 [:memory:]
  device avg_duration
  <chr>    <dbl>
1 laptop    376.
2 mobile    337.
3 tablet    354.

```

2.5 Show Query

If you want to view the SQL query generated in the above step, use `show_query()` or `explain()`.

```

avg_time <-
ecom2 %>%

```

```

group_by(device) %>%
  summarise(avg_duration = mean(duration))

dplyr::show_query(avg_time)
## <SQL>
## SELECT `device`, AVG(`duration`) AS `avg_duration`
## FROM `ecom`
## GROUP BY `device`

dplyr::explain(avg_time)
## <SQL>
## SELECT `device`, AVG(`duration`) AS `avg_duration`
## FROM `ecom`
## GROUP BY `device`
##
## <PLAN>
##   id parent notused          detail
## 1  6      0      115          SCAN ecom
## 2  8      0         0 USE TEMP B-TREE FOR GROUP BY

```

2.6 Collect Data

Now, some interesting facts. When working with databases, **dplyr** never pulls data into R unless you explicitly ask for it. In the previous example, dplyr will not do anything until you ask for the `avg_time` data. It generates the SQL and only pulls down a few rows when you try to print `avg_time`. So how do we pull all the data and store it for further analysis? `collect()` will pull all the data and store it in a tibble and you can use it for any further analysis.

```

dplyr::collect(avg_time)

# A tibble: 3 x 2
  device avg_duration
  <chr>      <dbl>
1 laptop      376.
2 mobile      337.
3 tablet      354.

```

It is good practice to close the connection when you are done:

```
dbDisconnect(con)
```

3 SQL Basics

3.1 Introduction

In this chapter, we will learn to:

- select
 - single column

- multiple columns
- distinct values in a column
- limit the number of records returned
- handle NULL values
- and filter columns using the following operators
 - WHERE
 - AND, or & NOT
 - BETWEEN
 - IN
 - LIKE

We will use the following R packages:

- DBI
- RSQLite
- dbplyr

All the data sets used in this chapter can be found [here](#) and code can be downloaded from [here](#) or run from `code/ch3-sqlbasics.R` in this repo.

3.2 Set Up

In order to be able to use the `ecom` data set for our case study, we will read it using the **readr** package and then copy it to an in memory database using the `copy_to()` function from **dplyr**.

```
# Canonical ecom: 1000 rows x 8 cols (referrer, device, bouncers, n_visit,
# n_pages, duration, country, purchase), from the local data/web.csv snapshot.
web_path <- if (file.exists("data/web.csv")) "data/web.csv" else "https://raw.githubusercontent.com/rsquaredacademy/datasets/master/web.csv"
ecom <- readr::read_csv(web_path, show_col_types = FALSE)
ecom <- dplyr::select(ecom, referrer, device, bouncers, n_visit, n_pages, duration,
country, purchase)
con <- DBI::dbConnect(RSQLite::SQLite(), ":memory:")
dplyr::copy_to(con, ecom)
```

3.3 Select Columns

The SQL SELECT statement is used to fetch the data from a database table.

3.3.1 Syntax

Below is the basic syntax of the SELECT statement.

```
SELECT column1, column2, columnN FROM table_name;
```

Here, column1, column2... are the fields of a table whose values you want to fetch. If you want to fetch all the fields, use the following syntax.

```
SELECT * FROM table_name;
```

3.3.2 Select Single Column

Let us begin by selecting the `device` field from the `ecom` table.

```
SELECT device FROM ecom
```

device
laptop
tablet
laptop
tablet
mobile
laptop
mobile
mobile
mobile
mobile

3.3.3 Select Multiple Columns

Select the following fields from the `ecom` table:

- `referrer`
- `device`
- `purchase`

```
SELECT referrer, device, purchase FROM ecom
```

referrer	device	purchase
google	laptop	0
yahoo	tablet	0
direct	laptop	0
bing	tablet	1
yahoo	mobile	0
yahoo	laptop	0
yahoo	mobile	0
direct	mobile	0
bing	mobile	0
google	mobile	0

3.3.4 Select All Columns

Select all the fields from the `ecom` table.

```
SELECT * FROM ecom
```

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
google	laptop	1	10	1	693	Czech Republic	0
yahoo	tablet	1	9	1	459	Yemen	0
direct	laptop	1	0	1	996	Brazil	0
bing	tablet	0	3	18	468	China	1
yahoo	mo- bile	1	9	1	955	Poland	0
yahoo	laptop	0	5	5	135	South Africa	0
yahoo	mo- bile	1	10	1	75	Bangladesh	0
direct	mo- bile	1	10	1	908	Indonesia	0
bing	mo- bile	0	3	19	209	Netherlands	0
google	mo- bile	1	6	1	208	Czech Republic	0

3.4 Limit

If you have a large table with thousands of rows, returning all the records will take time. Use `LIMIT` to specify the number of records to return.

```
SELECT * FROM ecom limit 10
```

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
google	laptop	1	10	1	693	Czech Republic	0
yahoo	tablet	1	9	1	459	Yemen	0
direct	laptop	1	0	1	996	Brazil	0
bing	tablet	0	3	18	468	China	1
yahoo	mo- bile	1	9	1	955	Poland	0
yahoo	laptop	0	5	5	135	South Africa	0

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
yahoo	mo- bile	1	10	1	75	Bangladesh	0
direct	mo- bile	1	10	1	908	Indonesia	0
bing	mo- bile	0	3	19	209	Netherlands	0
google	mo- bile	1	6	1	208	Czech Republic	0

3.5 Distinct

A column in a table may often contain many duplicate values; and we might be interested only in the distinct/unique values. In such cases, we can use the `SELECT DISTINCT` statement to return only distinct values.

```
SELECT distinct referrer FROM ecom
```

```
referrer
-----
google
yahoo
direct
bing
social
```

3.6 Filter

Now that we know how to select columns, let us focus on filtering data. In SQL, the `WHERE` keyword is used to extract only those records that fulfill a specified condition. Data filter based on both text and numeric values in a table. Below are a few comparison operators we can use:

- `=` equal
- `<>` not equal
- `<` less than
- `>` greater than
- `<=` less than or equal to
- `>=` greater than or equal to

The following SQL statement filters all rows from the `ecom` table where the `duration` field is greater than 300.

```
SELECT *
FROM ecom
WHERE duration > 300
```

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
google	laptop	1	10	1	693	Czech Republic	0
yahoo	tablet	1	9	1	459	Yemen	0
direct	laptop	1	0	1	996	Brazil	0
bing	tablet	0	3	18	468	China	1
yahoo	mo- bile	1	9	1	955	Poland	0
direct	mo- bile	1	10	1	908	Indonesia	0
direct	laptop	1	9	1	738	Jamaica	0
direct	mo- bile	0	9	14	406	Ireland	1
bing	laptop	1	1	1	995	United States	0
bing	tablet	0	5	16	368	Peru	1

Let us filter data based on a text value. In the following example, we filter all rows from the `ecom` table where the `device` used is `mobile`.

```
SELECT *
FROM ecom
WHERE device = 'mobile'
```

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
yahoo	mo- bile	1	9	1	955	Poland	0
yahoo	mo- bile	1	10	1	75	Bangladesh	0
direct	mo- bile	1	10	1	908	Indonesia	0
bing	mo- bile	0	3	19	209	Netherlands	0

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
google	mo- bile	1	6	1	208	Czech Republic	0
direct	mo- bile	0	9	14	406	Ireland	1
yahoo	mo- bile	0	7	1	19	France	0
google	mo- bile	1	5	1	147	Brazil	0
bing	mo- bile	0	0	7	196	Russia	0
google	mo- bile	1	10	1	338	Russia	0

3.7 And, Or & Not

The WHERE clause can be combined with other operators such as

- AND - displays a record if all the conditions separated by AND is TRUE
- OR - displays a record if any of the conditions separated by OR is TRUE
- NOT - displays a record if the condition(s) is NOT TRUE

to filter data based on more than one condition or to create more complex conditions.

In the following example, we filter all the rows from the `ecom` table where `n_visit` (visit count) is greater than 3 and `duration` (time spent on the site) is greater than 100. We use AND to create multiple conditions.

```
SELECT *
FROM ecom
WHERE n_visit > 3 AND duration > 100
```

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
google	laptop	1	10	1	693	Czech Republic	0
yahoo	tablet	1	9	1	459	Yemen	0
yahoo	mo- bile	1	9	1	955	Poland	0
yahoo	laptop	0	5	5	135	South Africa	0

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
direct	mo- bile	1	10	1	908	Indonesia	0
google	mo- bile	1	6	1	208	Czech Republic	0
direct	laptop	1	9	1	738	Jamaica	0
direct	tablet	0	6	12	132	Estonia	0
direct	mo- bile	0	9	14	406	Ireland	1
bing	tablet	0	5	16	368	Peru	1

In the next example, we will use both AND & OR. Our goal is to filter all rows from the `ecom` table that follow the below conditions:

- `n_visit` (visit count) is either equal to 3 or 5
- `device` used to visit the website is either `mobile` or `tablet`

```
SELECT *
FROM ecom WHERE (n_visit = 5 OR n_visit = 3)
AND (device = 'mobile' OR device = 'tablet')
```

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
bing	tablet	0	3	18	468	China	1
bing	mobile	0	3	19	209	Netherlands	0
yahoo	tablet	0	5	8	80	Philippines	0
bing	tablet	0	5	16	368	Peru	1
google	mobile	1	5	1	147	Brazil	0
yahoo	mobile	1	3	1	700	Canada	0
bing	tablet	1	5	1	831	Iran	0
social	tablet	0	3	12	324	China	0
yahoo	tablet	0	3	5	140	Latvia	0
yahoo	tablet	1	3	1	332	France	0

3.8 BETWEEN

The BETWEEN operator selects values within a given range and is inclusive: begin and end values are included. The values can be numbers, text, or dates. In the following example, we filter rows

from the `ecom` table where the visit count is between 1 and 3, and the `device` used to visit the website is `mobile`.

```
SELECT *
FROM ecom
WHERE n_visit BETWEEN 1 AND 3 AND device = 'mobile'
```

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
bing	mobile	0	3	19	209	Netherlands	0
direct	mobile	1	2	1	501	El Salvador	0
bing	mobile	0	1	1	25	Ireland	0
yahoo	mobile	1	3	1	700	Canada	0
direct	mobile	0	1	13	234	Indonesia	0
social	mobile	0	1	2	44	France	0
bing	mobile	0	3	18	288	Portugal	1
yahoo	mobile	1	1	1	73	Indonesia	0
google	mobile	0	2	3	69	Bahamas	0
direct	mobile	1	1	1	423	Mexico	0

3.9 IN

The `IN` operator allows us to specify multiple values in a `WHERE` clause. It is a shorthand for multiple `OR` conditions. In the below example, we filter rows from the `ecom` table where the visit count is either 2 or 4 or 6 or 8 or 10. Instead of using multiple `OR` conditions, we use the `IN` operator.

```
SELECT *
FROM ecom
WHERE n_visit IN (2, 4, 6, 8, 10)
```

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
google	laptop	1	10	1	693	Czech Republic	0
yahoo	mobile	1	10	1	75	Bangladesh	0
direct	mobile	1	10	1	908	Indonesia	0
google	mobile	1	6	1	208	Czech Republic	0
direct	tablet	0	6	12	132	Estonia	0

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
social	tablet	0	2	1	28	Namibia	0
direct	laptop	1	2	1	384	China	0
social	laptop	1	4	1	850	Ukraine	0
direct	tablet	0	2	19	342	Japan	1
google	mo- bile	1	10	1	338	Russia	0

3.10 IS NULL

A field with a NULL value is a field with no value. If a field in a table is optional, it is possible to insert a new record or update a record without adding a value to this field. Then, the field will be saved with a NULL value. In the next example, we filter all rows from the `ecom` table where the `device` column has NULL values.

```
SELECT *
FROM ecom
WHERE device IS NULL
```

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
----------	--------	----------	---------	---------	----------	---------	----------

3.11 LIKE

The LIKE operator is used to search for a specific pattern in a column. There are two wildcards used in conjunction with the LIKE operator:

- `%` : represents zero, one, or multiple characters
- `_` : represents a single character

In the following example, we filter all rows from the `ecom` table where the name of the `country` starts with `P`. We use `%` after `P` to indicate that it can be followed by any number or type of characters.

```
SELECT *
FROM ecom
WHERE country LIKE 'P%'
```

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
yahoo	mobile	1	9	1	955	Poland	0
yahoo	tablet	0	5	8	80	Philippines	0
bing	tablet	0	5	16	368	Peru	1
bing	laptop	1	0	1	456	Portugal	0

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
yahoo	tablet	1	9	1	706	Philippines	0
direct	tablet	1	8	1	814	Philippines	0
social	laptop	0	5	16	176	Panama	0
yahoo	laptop	1	4	1	928	Peru	0
yahoo	tablet	0	3	11	176	Philippines	0
google	laptop	1	9	1	869	Poland	0

Let us look at another example where we filter all rows from the `ecom` table where the name of the country should follow the below conditions:

- name can start with any character
- the second character must be `o`
- it can have any type or number of characters after the second character

```
SELECT *
FROM ecom
WHERE country LIKE '_o%'
```

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
yahoo	mo- bile	1	9	1	955	Poland	0
yahoo	laptop	0	5	5	135	South Africa	0
social	tablet	0	7	10	290	Colombia	1
yahoo	mo- bile	0	8	9	225	Colombia	0
direct	laptop	1	8	1	658	Colombia	0
bing	laptop	1	0	1	456	Portugal	0
yahoo	tablet	0	7	11	264	Somalia	0
google	tablet	0	1	19	304	Colombia	0
google	laptop	1	9	1	869	Poland	0
social	tablet	1	1	1	566	Portugal	0

```
dbDisconnect(con)
```

3.12 Exercises

1. List the `referrer` and `device` of 5 visits with more than 8 page views (`n_visit > 8`), ordered by longest `duration` first.
2. How many rows are from `tablet` devices in countries starting with `P`?

3. Which distinct `device` values appear among bouncers (`bouncers = 1`)?

Worked solutions in `solutions/sql-basics.md` (attempt first).

4 SQL Advanced

4.1 Introduction

In this chapter, we will learn to

- aggregate data
- order data
- and group data

We will use the following R packages:

- DBI
- RSQLite
- dbplyr

All the data sets used in this chapter can be found here and code can be downloaded from here or run from `code/ch4-sqladvanced.R` in this repo.

4.2 Set Up

In order to be able to use the `ecom` data set for our case study, we will read it using the `readr` package and then copy it to an in memory database using the `copy_to()` function from `dbplyr`.

```
# Canonical ecom: 1000 rows x 8 cols (referrer, device, bouncers, n_visit,
# n_pages, duration, country, purchase), from the local data/web.csv snapshot.
web_path <- if (file.exists("data/web.csv")) "data/web.csv" else "https://raw.githubusercontent.com/rsquaredacademy/datasets/master/web.csv"
ecom <- readr::read_csv(web_path, show_col_types = FALSE)
ecom <- dplyr::select(ecom, referrer, device, bouncers, n_visit, n_pages, duration,
country, purchase)
con <- DBI::dbConnect(RSQLite::SQLite(), ":memory:")
dplyr::copy_to(con, ecom)
```

4.3 Aggregate

Let us combine the aggregate statements with WHERE statement to filter data.

- `SUM()` : returns the total sum of a numeric column

```
SELECT SUM(n_visit) FROM ecom
```

SUM(n_visit)
4972

```
SELECT SUM(n_visit)
FROM ecom
WHERE n_visit > 5
```

SUM(n_visit)
3574

- AVG() : returns the average value of a numeric column

```
SELECT AVG(n_visit) FROM ecom
```

AVG(n_visit)
4.972

```
SELECT AVG(n_visit)
FROM ecom
WHERE country LIKE 'P%'
```

AVG(n_visit)
5.079137

- MAX() : returns the largest value of the selected column

```
SELECT MAX(n_visit) FROM ecom
```

MAX(n_visit)
10

```
SELECT MAX(n_visit)
FROM ecom
WHERE device = 'tablet'
```

MAX(n_visit)
10

- MIN() : returns the smallest value of the selected column

```
SELECT MIN(n_visit) FROM ecom
```

MIN(n_visit)
0

```
SELECT MIN(n_visit)
FROM ecom
WHERE duration BETWEEN 600 AND 900
```

MIN(n_visit)
0

4.4 Alias

SQL aliases are used to give a table, or a column in a table, a temporary name. They are often used to make column names more readable. An alias only exists for the duration of the query. Below are a few examples:

```
SELECT AVG(n_visit) AS avg_mobile
FROM ecom
WHERE device = 'mobile'
```

avg_mobile
5.479651

```
SELECT MAX(n_visit) AS max_visit
FROM ecom
```

max_visit
10

```
SELECT MIN(duration) AS min_duration
FROM ecom
```

min_duration
10

4.5 Order By

The ORDER BY keyword is used to sort the records in ascending or descending order. By default, the records are sorted in ascending order. Use the DESC keyword if you want to sort the records in descending order,

```
SELECT *
FROM ecom
ORDER BY country
```

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
social	laptop	0	8	2	60	Afghanistan	0
yahoo	laptop	0	10	18	180	Afghanistan	0
social	laptop	1	2	1	274	Afghanistan	0
direct	tablet	1	10	1	682	Afghanistan	0

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
direct	tablet	0	5	18	414	Afghanistan	1
bing	tablet	0	7	2	34	Afghanistan	0
direct	laptop	0	6	6	96	Albania	0
google	laptop	1	5	1	304	Albania	0
bing	mobile	0	2	2	54	Albania	0
yahoo	laptop	0	3	1	23	Albania	0

```
SELECT *
FROM ecom
ORDER BY duration
```

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
yahoo	tablet	1	5	1	10	Poland	0
social	laptop	1	1	1	10	Finland	0
yahoo	laptop	0	0	1	12	Indonesia	0
social	mobile	1	2	1	12	Botswana	0
direct	laptop	0	0	1	13	Czech Republic	0
direct	mobile	0	10	1	17	China	0
yahoo	tablet	0	4	1	17	Poland	0
yahoo	mobile	0	7	1	19	France	0
bing	laptop	0	5	1	20	Thailand	0
yahoo	mobile	0	6	1	21	Russia	0

```
SELECT *
FROM ecom
ORDER BY n_visit DESC
```

referrer	device	bouncers	n_visit	n_pages	duration	country	purchase
google	laptop	1	10	1	693	Czech Republic	0

referrer	device	bounc- ers	n_visit	n_pages	duration	country	purchase
yahoo	mo- bile	1	10	1	75	Bangladesh	0
direct	mo- bile	1	10	1	908	Indonesia	0
google	mo- bile	1	10	1	338	Russia	0
google	mo- bile	1	10	1	479	Cameroon	0
direct	mo- bile	0	10	1	17	China	0
direct	tablet	0	10	3	57	China	0
social	mo- bile	0	10	11	242	Argentina	1
direct	mo- bile	1	10	1	675	Indonesia	0
google	tablet	1	10	1	86	China	0

4.6 Group By

The GROUP BY statement is used with aggregate functions (COUNT, MAX, MIN, SUM, AVG) to group the result by one or more columns.

```
SELECT device,
count(*) AS visits
FROM ecom
GROUP BY device
ORDER by visits DESC
```

device	visits
mobile	344
tablet	331
laptop	325

```
SELECT device, MAX(duration) AS max_duration
FROM ecom
GROUP BY device
ORDER by max_duration DESC
```

device	max_duration
tablet	999
laptop	997
mobile	994

`dbDisconnect(con)`

4.7 Exercises

1. Compute the average pages per visit (`AVG(n_pages)`) for each `referrer`, ordered highest average first. Which referrer tops the list?
2. What is the longest `duration` among visits that made a purchase (`purchase = 1`)?
3. Among `mobile` visits only, count visits by `country` and show the top 5. Which country is first?

Worked solutions in `solutions/sql-advanced.md` (attempt first).

5 JOINS

5.1 Introduction

In this chapter, we will learn to:

- combine rows from two tables with `INNER JOIN`
- keep unmatched rows with `LEFT JOIN`
- avoid the SQLite `RIGHT / FULL OUTER JOIN` trap
- do the same joins from R with `dbplyr inner_join()` / `left_join()`
- filter with `semi_join()` and `anti_join()`

We will use the following R packages:

- DBI
- RSQLite
- dbplyr

All the data sets used in this chapter can be found here and code can be run from `code/ch5-joins.R` in this repo.

5.2 Set Up

So far we have worked with a single table. Real databases keep related data in separate tables. Our `data/ecom.sqlite` file holds two small related tables besides `ecom`:

- `customers`: 20 rows, one per customer (`customer_id` primary key, `country`)
- `orders`: 20 rows, one per order (`order_id` primary key, `customer_id` foreign key into `customers`, `amount`)

Six customers (4, 16–20) have placed no orders, and one order points at a `customer_id` (99) that does not exist. These deliberate mismatches make the difference between join types visible.

Let us connect directly to the SQLite file:

```
con <- DBI::dbConnect(RSQLite::SQLite(), "data/ecom.sqlite")
```

A quick look at both tables:

```
SELECT * FROM customers LIMIT 5
```

customer_id	country
1	United States
2	United Kingdom
3	Germany
4	France
5	Canada

```
SELECT * FROM orders LIMIT 5
```

order_id	customer_id	amount
101	1	459.11
102	1	469.80
103	2	157.35
104	3	418.61
105	3	328.04

5.3 INNER JOIN

INNER JOIN returns only the rows that match in **both** tables. Customers without orders, and orders without customers, are dropped.

```
SELECT c.customer_id, c.country, o.order_id, o.amount
FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
ORDER BY c.customer_id, o.order_id
LIMIT 6
```

customer_id	country	order_id	amount
1	United States	101	459.11
1	United States	102	469.80
2	United Kingdom	103	157.35
2	United Kingdom	118	76.39
3	Germany	104	418.61
3	Germany	105	328.04

How many of the 20 orders survive? Nineteen – order 120 belongs to the non-existent customer 99 and is excluded:

```
SELECT COUNT(*) AS n FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
```

n
19

5.4 LEFT JOIN

LEFT JOIN keeps **every** row from the left table and fills **NULL** where the right table has no match. This is how you find customers who never ordered:

```
SELECT c.customer_id, c.country, o.order_id, o.amount
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL
ORDER BY c.customer_id
```

customer_id	country	order_id	amount
4	France	NA	NA
16	Italy	NA	NA
17	Portugal	NA	NA
18	Greece	NA	NA
19	Ireland	NA	NA
20	Belgium	NA	NA

The full left join has 25 rows: 19 matched order rows plus one **NULL** row for each of the 6 customers without orders.

5.5 The SQLite RIGHT / FULL Trap

Some tutorials teach **RIGHT JOIN** or **FULL OUTER JOIN**. Do not use them with SQLite: the SQLite versions bundled with SQLite (including recent ones) reject both with a syntax error such as **near "RIGHT": syntax error**. Postgres and other production databases accept them, so queries written against SQLite will fail silently only when you move them.

The portable workaround is to swap the tables and use **LEFT JOIN**:

- instead of **A RIGHT JOIN B**, write **B LEFT JOIN A**
- instead of **A FULL OUTER JOIN B**, union a **LEFT JOIN** in each direction

When in doubt, **LEFT JOIN** is the only outer join you need for this book.

5.6 JOINS with dbplyr

The same joins work from R without writing SQL. Reference both tables and use the dplyr join verbs:

```
customers <- dplyr::tbl(con, "customers")
orders <- dplyr::tbl(con, "orders")

inner_join(customers, orders, by = "customer_id")

# A query: ?? x 4
# Database: sqlite 3.53.3 [/home/runner/work/rdbsql/rdbsql/data/ecom.sqlite]
  customer_id country      order_id amount
      <int> <chr>         <int> <dbl>
1           1 United States    101  459.
2           1 United States    102  470.
3           2 United Kingdom   103  157.
4           2 United Kingdom   118   76.4
5           3 Germany          104  419.
6           3 Germany          105  328.
7           3 Germany          106  269.
8           5 Canada           107  374.
9           5 Canada           119  248.
10          6 Australia         108   84.6
11          7 Brazil            109  335.
12          8 Japan             110  358.
13          9 Mexico            111  240.
14         10 Netherlands        112  365.
15         11 Sweden             113  469.
16         12 Norway             114  143.
17         13 Denmark            115  242.
18         14 Finland            116  471.
19         15 Poland             117  490.
```

Use `show_query()` to see the SQL dbplyr generates – it is the `INNER JOIN` you just wrote by hand:

```
q_inner <- inner_join(customers, orders, by = "customer_id")
dplyr::show_query(q_inner)
## <SQL>
## SELECT `customers`.*, `order_id`, `amount`
## FROM `customers`
## INNER JOIN `orders`
##   ON (`customers`.`customer_id` = `orders`.`customer_id`)
```

The left join works the same way:

```
q_left <- left_join(customers, orders, by = "customer_id")
dplyr::show_query(q_left)
```

```
## <SQL>
## SELECT `customers`.*, `order_id`, `amount`
## FROM `customers`
## LEFT JOIN `orders`
##   ON (`customers`.`customer_id` = `orders`.`customer_id`)
```

5.7 semi_join and anti_join

Two filtering joins have no direct SQL keyword but are everyday tools:

- `semi_join()` keeps rows of `customers` that **have** a match in `orders` (14 customers). The generated SQL uses `WHERE EXISTS`.
- `anti_join()` keeps rows of `customers` that **lack** a match in `orders` (the same 6 customers from the `LEFT JOIN` check above). The generated SQL uses `WHERE NOT EXISTS`.

```
q_semi <- semi_join(customers, orders, by = "customer_id")
dplyr::show_query(q_semi)
## <SQL>
## SELECT `customers`.*
## FROM `customers`
## WHERE EXISTS (
##   SELECT 1 FROM `orders`
##   WHERE (`customers`.`customer_id` = `orders`.`customer_id`)
## )
dplyr::collect(q_semi)
## # A tibble: 14 x 2
##   customer_id country
##   <int> <chr>
## 1         1 United States
## 2         2 United Kingdom
## 3         3 Germany
## 4         5 Canada
## 5         6 Australia
## 6         7 Brazil
## 7         8 Japan
## 8         9 Mexico
## 9        10 Netherlands
## 10       11 Sweden
## 11       12 Norway
## 12       13 Denmark
## 13       14 Finland
## 14       15 Poland

q_anti <- anti_join(customers, orders, by = "customer_id")
dplyr::show_query(q_anti)
## <SQL>
## SELECT `customers`.*
```

```
## FROM `customers`
## WHERE NOT EXISTS (
##   SELECT 1 FROM `orders`
##   WHERE (`customers`.`customer_id` = `orders`.`customer_id`)
## )
dplyr::collect(q_anti)
## # A tibble: 6 x 2
##   customer_id country
##   <int> <chr>
## 1         4 France
## 2        16 Italy
## 3        17 Portugal
## 4        18 Greece
## 5        19 Ireland
## 6        20 Belgium
```

Rule of thumb: use `semi_join()` / `anti_join()` when you want to filter one table by membership in another without adding the other table's columns.

```
dbDisconnect(con)
```

5.8 Exercises

1. List the `customer_id` and `country` of customers who placed no orders. Write it both as a `LEFT JOIN ... IS NULL` query and as an `anti_join()`.
2. Compute total order `amount` per customer (highest first) with an `INNER JOIN + GROUP BY`. Which customer spent the most?
3. Find the orphan order: which `order_id` points at a non-existent customer?

Worked solutions in `solutions/joins.md` (attempt first).

6 DBI Cheat Sheet

One-page reference: connect, query, disconnect. Printable PDF companion to this book (CC BY-NC-SA 4.0).

6.1 Connect, query, disconnect

```
library(DBI)
con <- dbConnect(RSQLite::SQLite(), ":memory:") # or duckdb::duckdb()
dbListTables(con)                             # what is in the database?
dbReadTable(con, "ecom")                       # whole table (small only!)
dbGetQuery(con, "SELECT * FROM ecom LIMIT 10")  # SQL, first rows
res <- dbSendQuery(con, "SELECT * FROM ecom")   # big tables: fetch in batches
x <- dbFetch(res, n = 100)
dbClearResult(res)                             # always clear when done
dbDisconnect(con)                              # always disconnect
```

6.2 Write data

```
dbWriteTable(con, "trial", df)           # create (overwrite / append args)
dbExecute(con, "INSERT INTO trial (x) VALUES (1)") # statements, no cleanup needed
dbRemoveTable(con, "trial")              # delete table
```

6.3 Inspect a query

`dbHasCompleted()` `dbGetStatement()` `dbGetRowCount()` `dbGetRowsAffected()` `dbColumnInfo()`
— status, SQL, counts, column types of a `dbSendQuery()` result.

6.4 dbplyr in 4 verbs

```
e <- dplyr::tbl(con, "ecom")           # lazy reference, pulls nothing
e |> dplyr::filter(duration > 300)     # WHERE
e |> dplyr::group_by(device) |>        # GROUP BY + aggregates
  dplyr::summarise(avg = mean(duration))
dplyr::show_query(q)                   # see the generated SQL
dplyr::collect(q)                      # pull results into a tibble
inner_join(a, b) / left_join(a, b)     # JOINS; semi/anti_join filter
```